
MLET: Selective Neural Residual Correction for Spatial Evapotranspiration

Marco Trotta



Irrigant, Idaho, USA

m@irrigant.xyz

September 22, 2026

Abstract

A neural correction of a satellite estimate must decide both what to predict and when its correction is useful. We study this decision as selective residual regression under spatial transfer and input corruption. Using 7,923 OpenET station-days from 85 stations, we compare ensemble disagreement, input-support distance, and learned relative benefit. Selectors use spatially cross-fitted neural predictions; a separate evaluation withholds both locations and later years. The experiment retains a fixed satellite fallback and scores every test observation. A natural humidity anomaly motivates a separate feature-omission arm and controlled test-input faults. After VPD omission, a tenfold wind-input fault raises residual-network station MAE from 0.829 to 1.415 mm day⁻¹ on 7,873 matched observations. Support screening yields 0.857, but largely through fallback: it accepts only 1.74% of the station-weighted population. At matched 50% acceptance, its advantage over disagreement is inconclusive. However, the learned benefit ranking is worse than the support ranking by 0.243 mm day⁻¹, with paired interval [0.110, 0.412], under that same fault. The later-year evaluation does not confirm this ordering, and natural-test gains do not transfer uniformly to croplands. We derive the baseline-relative risk objective, preserve all negative controls, and distinguish screening from ranking improvements. The result isolates a failure of cross-fitted benefit selection under a specified input fault and separates correction ranking from fallback frequency.

Keywords: selective regression; neural residual learning; model deferral; input corruption; evapotranspiration

1 Introduction

Machine Learning Evapotranspiration (MLET) combines satellite estimates, meteorological covariates, and auditable environmental prediction. A satellite model already provides an actual-ET estimate where direct flux measurements are unavailable. A neural residual model can correct its error, but that correction can also damage a useful estimate. The technical question is therefore conditional: *when should the system apply the learned correction at an unseen location?*

This question connects environmental modeling to regression with deferral. An uncertainty score estimates reliability of a model, whereas the correction decision compares that model with an available fallback. Low neural error does not necessarily mean improvement over the satellite estimate. Moreover, a learned selector faces its own distribution shift when local weather inputs or their support change.

We investigate this narrow problem using matched neural predictions and a fixed OpenET fallback. We compare a disagreement threshold, a training-support threshold, and a learned absolute-loss benefit selector. A crossed experiment separates geographic withholding, later years, and omission of a previously identified anomalous covariate. Paired input faults then isolate selection behavior while holding labels and neural parameters fixed. The contribution is a reproducible empirical study of selector transfer, supported by an explicit loss objective and matched-acceptance diagnostics.

Agricultural use requires a separate population check: success across mixed land covers need not imply success on croplands. We therefore report cropland transfer alongside the full archive and retain the satellite estimate as a deployable fallback concept. The evaluation is retrospective; it does not test operational availability or irrigation interventions.

The study makes three connected empirical contributions. First, it evaluates selection by improvement over the fallback, with spatially cross-fitted targets and complete-system loss. Second, paired input faults separate neural error, rejection frequency, and correction ranking while holding the fitted predictor fixed. Third, matched acceptance exposes an inconclusive uncertainty comparison and a specific benefit-selector failure that later-year data do not confirm. Appendices G and H retain the original MLET architecture and one reference-ETo feasibility case as historical context.

2 Related work

Mao et al. (2024) formulate regression with multi-expert deferral, including two-stage methods and consistency analysis. This is the closest formal setting: MLET chooses between a neural correction and a satellite predictor. We use a direct, cross-fitted loss-difference regressor as a fixed baseline. We do not reproduce their surrogate losses or claim their consistency bounds for this unbounded absolute-loss experiment.

Sokol et al. (2024) study conformalized selective regression and connect uncertainty with coverage. Our objective includes the fallback’s error on rejected observations, rather than report accuracy only on retained predictions. We claim no conformal coverage guarantee under geographic shift. We use seed disagreement following Lakshminarayanan et al. (2016). Its behavior can be unreliable outside training support (de Mathelin et al., 2023).

Applicability-domain methods relate environmental prediction reliability to covariate support (Meyer and Pebesma, 2020). GeoQ uses geometry, cross-fitting, and conditional quantile increments to estimate scientific-surrogate errors (Nguyen et al., 2026). Our distance control is simpler; it tests whether support screening changes the interpretation of a learned selector. Work on ReLU extrapolation further motivates separating interpolation from unsupported input directions (Xu et al., 2020). Graph neural processes demonstrate environmental prediction at sparse sensor locations (Hu et al., 2023). Their graph and temporal context differs from the sparse satellite-date regression studied here.

Volk et al. (2024) evaluate OpenET against a public collection of measured ET. The associated model and measurement archives provide the observations used here (Volk et al., 2023b,a). Our reuse of that archive does not constitute an independent evaluation of upstream OpenET model development. The held-out partitions apply to the additional correction models trained in this study.

Machine learning correction of ET is already an established research direction. Hascoet et al. (2022) learn corrections using water-cycle closure supervision. Rozanov et al. (2025) combine physical features, tree models, and neural models to estimate ET across the U.S. Midwest. Shi (2024) study domain-adversarial learning for geographical ET transfer. Recent work also studies residual correction of physical ET estimates and physically constrained learning (Dobrokhotov et al., 2026; Wei et al., 2026). These studies motivate strong correction baselines; they do not establish that a particular correction transfers in the present archive. OpenET also documents land-cover-specific biases, including forest overprediction (OpenET, 2026).

Structured validation must reflect the dependence present in environmental data (Roberts et al., 2017). Nearby towers can share weather, management, and landscape conditions despite having different station identifiers. We therefore complement station withholding with proximity groups. We also distinguish site-year grouping from simultaneously excluding a station and all later test years. The latter prevents training on earlier observations of a test station or on later observations from other stations.

Reference ET is a separate quantity defined for a standardized surface. Kukul et al. (2026) question replacing its specified calculation with a learned approximation. Our target is measured actual ET; reference ET is only an input. We do not infer crop water need, irrigation response, or reference-ET forecast skill from these results.

JEPA-Anything studies factorized predictive representations across domains (Cui et al., 2026). Dreamer studies learned dynamics for control (Hafner et al., 2025). They motivate explicit predictive structure and tests that isolate a proposed mechanism. They do not justify labeling a same-date scalar regression as an action-conditioned world model. The current archive lacks a matched irrigation-action sequence and a corresponding evaluation of intervention outcomes. Our decomposition concerns correction error, not identified causal factors.

3 Data and evaluation

3.1 A fixed retrospective cohort

We use the OpenET model archive, Zenodo record 10119477, and the processed flux archive, record 7636781. Both downloaded archives match the checksums in the existing MLET manifest. The exact station-date join produces 16,447 rows from 152 stations. Of these, 16,444 contain a measured, non-gap-filled corrected ET label. Requiring finite reference ET, temperature, vapor-pressure deficit (VPD), and wind leaves 7,923 observations from 85 stations. This reproduces the earlier complete-weather cohort exactly.

The selected rows span 415 station-years from 2001 through 2020. They are satellite validation dates, not complete daily sequences. The target is the archive’s `ET_corr`, which incorporates its energy-balance correction procedure. We retain negative target values and do not clip predictions. The complete-weather subset contains 32 cropland stations and 2,670 observations. The other stations represent grasslands, shrublands, forests, and wetland or riparian settings. Cropland classification does not establish irrigation status.

Let y_{st} denote corrected measured ET and o_{st} the OpenET ensemble at station s and date t . The input vector is

$$\begin{aligned} x_{st} &= [o_{st}, ET o_{st}, h_t, T_{st}, VPD_{st}, u_{st}], \\ h_t &= [\sin(2\pi d_t/365), \cos(2\pi d_t/365)]. \end{aligned} \quad (1)$$

Here d_t is day of year, T is average air temperature, and u is wind speed. The reference-ET covariate is gridMET ETo from the original joined table. We retain the earlier 365-day season encoding for reproducibility. The meteorological measurements and satellite estimates refer to the target date. Their operational release times are not reconstructed. Accordingly, this is retrospective estimation, not forecasting.

3.2 Three outer evaluations

All split assignments use seed 20260713. The first evaluation reproduces the original station-held-out 10-fold evaluation. No station contributes rows to both training and testing in one fold.

The primary evaluation constructs an undirected graph over the 85 stations. An edge connects stations whose great-circle distance is at most 10 km. Connected components define 63

proximity groups, including transitive connections. We allocate these groups to ten folds, keeping each component intact. All 7,923 observations receive exactly one held-out prediction per model. The 10 km threshold is fixed before this model comparison. It is a local-dependence check, not a test of transfer to an unseen climate region.

The joint evaluation uses the same proximity folds. Training additionally excludes dates from 2019 onward. Testing includes only dates from 2019 onward in held-out groups. This yields 649 observations from 27 stations and 24 groups. Only nine test stations are croplands, contributing 403 observations in seven groups. Comparisons within this evaluation use identical rows. Comparisons with the primary evaluation also reflect a changed test cohort.

3.3 Nested selection and inference

Inner selection uses three disjoint proximity-group folds drawn solely from the outer training set. In the joint evaluation, inner training ends before 2016 and validation starts in 2016. Outer test observations never determine scaling, stopping, regularization, or correction strength. We save the row identifiers for every inner and outer partition.

Our primary mean absolute error (MAE) gives equal weight to each test station:

$$\text{MAE}_{\text{station}}(f) = \frac{1}{S} \sum_{s=1}^S \frac{1}{n_s} \sum_{t=1}^{n_s} |y_{st} - f(x_{st})|. \quad (2)$$

We also report pooled MAE, pooled root mean square error (RMSE), and signed bias. Pooled MAE instead weights station s by n_s/N . These scores answer different questions when sampling density varies. Neither estimates an area-weighted agricultural error.

We report paired differences as baseline error minus candidate error. Positive differences favor the candidate. Intervals use 2,000 bootstrap samples of proximity components, with seed 20260922. Each sampled component contributes all its stations; repeated components retain their multiplicity. This preserves local clusters and avoids treating thousands of dates as independent replicates. The percentile intervals condition on the fitted cross-validation models. They omit retraining uncertainty and dependence between different proximity components. The exploratory comparisons have no multiplicity adjustment.

4 Neural correction models

4.1 Matched comparisons

We include the training median and the earlier crop-coefficient baseline in the accompanying results. Weather ridge uses reference ET, season, temperature, VPD, and wind. Combined ridge adds OpenET. Both standardize inputs using training means and standard deviations, then use ridge penalty $\alpha = 1$. Tuned ridge chooses $\alpha \in \{1, 10, 100, 1000\}$ by inner station-macro MAE. Affine calibration fits an intercept and slope to OpenET alone. Unchanged OpenET learns no parameters from this cohort.

Histogram gradient boosting supplies a nonlinear weather-only baseline. We also fit matched direct and residual boosting models with the complete input vector. Both use 150 it-

erations, learning rate 0.05, seven leaves, a 30-row minimum leaf size, and L2 regularization 10. Internal random-row early stopping is disabled.

The neural comparisons use two 32-unit ReLU layers and a scalar output. They have 1,345 trainable parameters with seven inputs. Direct and residual networks use identical architectures, optimizers, epoch counts, and seeds. We use Adam, learning rate 0.001, batch size 256, and 120 epochs. The scikit-learn L2 penalty parameter is 0.001. Inputs and training targets are standardized separately within each fit. No test-based stopping or architecture search occurs. We retain results for seeds 20260713, 20260714, and 20260715. The main neural predictions average the three seeds. Single-seed scores appear in Appendix D.

4.2 Residual prediction is a testable prior

A direct model predicts y . A residual model predicts $r = y - o$ and returns $o + g(x)$. The unchanged satellite estimate supplies an explicit fallback:

$$\hat{y}_\lambda = o + \lambda g(x), \quad 0 \leq \lambda \leq 1. \quad (3)$$

This construction does not make the residual predictable or transferable. For expressive model classes, direct and residual formulations can represent the same functions. Their practical differences arise from optimization, regularization, and inductive bias. They must therefore be compared with matched capacity and information.

Proposition 1 (Correction risk). *For a fixed correction g and any target distribution with finite second moments,*

$$\mathcal{R}(\lambda) - \mathcal{R}(0) = \lambda^2 B - 2\lambda A, \quad A = \mathbb{E}[rg], \quad B = \mathbb{E}[g^2], \quad (4)$$

where $\mathcal{R}(\lambda) = \mathbb{E}[(y - o - \lambda g)^2]$. If $B > 0$, the minimizing weight on $[0, 1]$ is $\lambda^* = \text{clip}(A/B, 0, 1)$. If $B = 0$, the correction has no effect almost surely.

Proof. Expand $(r - \lambda g)^2 - r^2$ and take expectations. The resulting convex quadratic has unconstrained minimizer A/B when $B > 0$. Projection onto $[0, 1]$ gives the constrained minimizer. $\square$

The identity distinguishes useful residual alignment, A , from correction energy, B . A full update helps squared error only if $2A > B$. A large correction without corresponding residual alignment damages the estimate. These moments also depend on the target population. Alignment measured across mixed land covers need not persist within croplands or later years.

The gated boosting model estimates A and B from inner out-of-group predictions, using equal station weights. It fixes $\hat{\lambda}$ before fitting the residual model on all outer training data. This is a familiar scalar shrinkage calculation. It has no distribution-free guarantee under station or temporal shift. It also optimizes squared loss, while the primary reporting metric is MAE. That mismatch is deliberate and must remain visible in the interpretation.

4.3 Why architecture and input support interact

Training standardization does not bound an unseen input. For linear prediction, a test perturbation δ changes the estimate by $\beta^\top D^{-1}\delta$, where D contains training scales. An extreme coordinate can therefore dominate the prediction even when all coefficients use ridge regularization. ReLU networks are piecewise affine and can also extrapolate along unsupported directions. A residual connection preserves the base estimate only when its learned update remains controlled.

A finite tree ensemble behaves differently for an input perturbation. Each tree returns a value from its finite set of leaves. For fixed trees h_j and positive learning rate η , its output satisfies

$$b + \eta \sum_j \min_{\ell} h_{j\ell} \leq b + \eta \sum_j h_j(x) \leq b + \eta \sum_j \max_{\ell} h_{j\ell}. \quad (5)$$

An extreme weather value therefore cannot create unbounded tree output for that fixed fit. This bound does not guarantee accurate predictions or prevent harmful training influence. It explains why input validity can interact with model family without establishing that one family is generally superior.

5 Selective neural residual correction

5.1 Predicting benefit differs from predicting error

Let $o(x)$ denote the available satellite estimate and $g(x)$ a fitted neural residual. A selector $a(x) \in \{0, 1\}$ returns

$$f_a(x) = o(x) + a(x)g(x). \quad (6)$$

Every query receives a prediction, including rejected corrections. Define the realized absolute-loss improvement as

$$D(x, y) = |y - o(x)| - |y - o(x) - g(x)|. \quad (7)$$

Positive D means that the correction helps. For a fixed fitted model, the complete-system risk satisfies

$$R(a) = R(o) - \mathbb{E}[a(X)D(X, Y)]. \quad (8)$$

Thus, the conditional optimal decision applies the correction when $\mu(x) = \mathbb{E}[D(X, Y) \mid X = x] > 0$. A selector that predicts small neural error need not predict positive improvement over o . This distinction is central to regression with deferral (Mao et al., 2024); we do not claim it as a new decision rule.

Consider one query with $y = o = 0$ and $g = 0.1$. A second query has $(y, o, g) = (10, 0, 9)$. The neural absolute errors are 0.1 and 1, but their improvements are -0.1 and 9. Rejecting the larger neural error retains the harmful correction and discards the helpful correction. Ensemble spread is further removed from this decision: it measures disagreement between fitted networks, not either realized error.

The selector’s target can be stated precisely. For $a^*(x) = \mathbf{1}\{\mu(x) > 0\}$,

$$R(a) - R(a^*) = \mathbb{E}[|\mu(X)|\mathbf{1}\{a(X) \neq a^*(X)\}]. \quad (9)$$

Conditioning Equation 8 on X proves the identity. Selection errors matter in proportion to the lost improvement, not their count alone. The triangle inequality also gives $|D| \leq |g|$. Consequently, an accepted correction can worsen absolute loss by

no more than its magnitude. This bound does not establish non-inferiority, and a support threshold does not change that limitation. These standard loss identities specify the experiment’s estimand rather than provide new consistency theory.

5.2 Spatial cross-fitting of the selector

The experiment fixes the three-seed residual ensemble from Section 4. Each outer training set produces three inner, geographically separated neural prediction sets. The joint evaluation also preserves its inner time cutoff. Only these inner predictions supply selector targets and thresholds. After selection, the neural ensemble is refitted on the complete outer training set. No outer target enters a fit, threshold, input scaler, or stopping decision. Figure 1 shows this separation.

For residual predictions g_1, g_2, g_3 , we use their mean $\bar{g}$ and sample spread

$$s_g(x) = \left[\frac{1}{2} \sum_{j=1}^3 (g_j(x) - \bar{g}(x))^2 \right]^{1/2}. \quad (10)$$

This is a small mean-only ensemble inspired by Lakshminarayanan et al. (2016); it is not their heteroscedastic likelihood model. A second score measures input support:

$$d_5(x) = \frac{1}{5} \sum_{z \in N_5(x)} \|S(x) - S(z)\|_2, \quad (11)$$

where S is the training-only standardizer and N_5 contains five training neighbors. The neighbor index is rebuilt inside every inner and outer training partition. Threshold calibration uses held-out inner distances, never distances from training examples to themselves. This simple control follows applicability-domain reasoning (Meyer and Pebesma, 2020); it does not reproduce weighted AOA or GeoQ.

The *Spread95* and *Support95* selectors accept scores below their station-weighted inner 95th percentiles. The *Gain* selector predicts Equation 7 with a fixed boosted-tree regressor. Its six inputs are $(\bar{g}, |\bar{g}|, s_g, \log(1 + d_5), o, \text{ETo})$. It uses 80 iterations, learning rate 0.05, seven leaves, minimum leaf size 50, L2 parameter 10, and no early stopping. Training weights give equal total weight to each station and are normalized to mean one. Gain accepts a correction only when its predicted improvement is positive. *SupportGain* requires both Gain and Support95 acceptance. The neural estimator remains identical across these selectors.

We include two controls. *Uniform* chooses one multiplier from $\{0, 0.25, 0.5, 0.75, 1\}$ using inner station MAE. *Clip* uses the unmodified fitted neural model but clips weather inputs to outer-training first and 99th percentiles at inference. Unchanged OpenET and the full residual ensemble define the endpoints. No selector hyperparameters are adjusted after viewing the outer results.

5.3 Natural transfer and controlled input faults

The selective experiment crosses two input specifications with two evaluations: proximity withholding and joint proximity-and-time withholding. The specifications contain seven archived inputs or six inputs without VPD. Each natural comparison retains every outer target. We score the complete

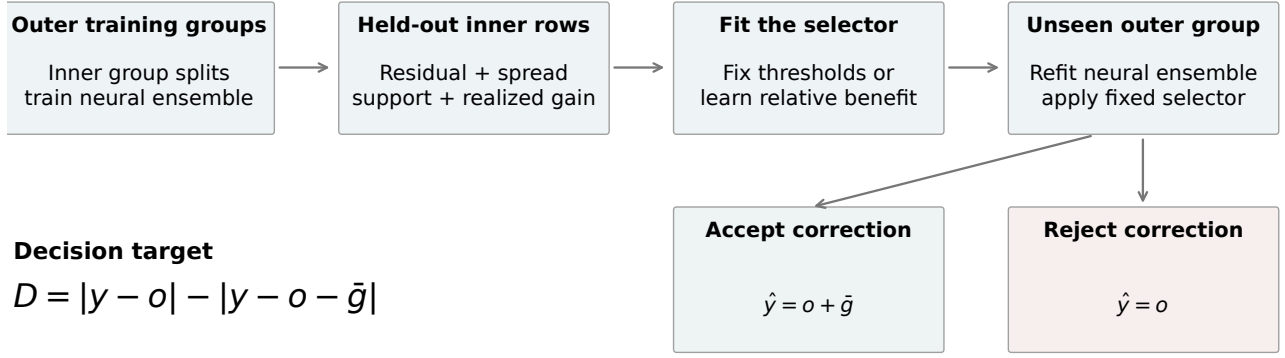


Figure 1. The selective residual experiment. Inner spatial cross-fitting supplies the selection data. Outer targets are used only for evaluation. Every rejected correction returns unchanged OpenET.

system, including fallback rows, and report croplands separately. The 95% inner thresholds do not impose 95% outer acceptance under distribution shift.

Three additional probes change test inputs separately: VPD multiplied by 10, wind multiplied by 10, and temperature increased by 20 °C. Their labels, satellite estimates, neural parameters, and selector parameters remain fixed. Only probe rows with nonnegative exported actual vapor pressure are eligible. Each fault is compared with its identical unmodified probe rows. The VPD probe is an exact negative control when VPD is omitted. These are synthetic input faults, not interventions on weather and not estimates of fault prevalence. The protocol specifies their magnitudes before the new run.

6 Selective transfer results

6.1 Screening and learning benefit behave differently

Table 1 reports every fixed selector on the natural test rows. Under proximity withholding with archived inputs, the full residual ensemble scores 1.099 station MAE. Spread95 and Support95 score 0.817 and 0.818 mm day⁻¹, respectively. They screen the known large neural failure, but neither improvement interval excludes zero against unchanged OpenET at 0.856. The learned Gain selector instead scores 1.126. Predicting relative benefit does not itself make that predictor reliable outside its calibration support.

After VPD omission, the full ensemble scores 0.835, Spread95 scores 0.832, and Support95 scores 0.838. Their improvement intervals remain inconclusive against OpenET. The simple Uniform control scores 0.816, with an exploratory improvement interval of [0.002, 0.093]. Its cropland error remains higher than the baseline. Thus, complexity in the selector is not necessary for the best observed point score in this comparison. Figure 2 retains all four evaluation settings.

The joint evaluation changes the interpretation again. For archived inputs, Uniform selects zero in every outer fold and exactly returns OpenET. The full ensemble nevertheless has lower observed station MAE on the later outer years, 0.766 versus 0.870. An earlier validation interval can reject corrections that later help on average. After VPD omission, the joint Gain, SupportGain, and full-model intervals all include zero.

These results give no general guarantee for the learned selector or the fallback rule.

6.2 Input support controls a specific neural fault

The unmodified probe subset contains 7,873 observations at 84 stations in 62 proximity groups. After VPD omission, its full-ensemble station MAE is 0.829. Multiplying wind inputs by ten raises that error to 1.415 mm day⁻¹. Spread95 scores 0.914 and Support95 scores 0.857, against 0.853 for unchanged OpenET. These are errors on identical probe rows, with identical fitted models and unchanged targets. The fault changes only the supplied wind coordinate.

The two thresholds spend different correction budgets. Spread95 accepts 19.96% of the station-weighted wind-fault population, whereas Support95 accepts 1.74%. On the natural no-VPD population, their acceptance rates are 95.74% and 96.84%. The small Support95 fault error therefore depends substantially on returning to the baseline. It is not evidence that its accepted neural predictions have uniformly superior accuracy.

The learned Gain selector accepts 56.95% under the wind fault and scores 1.261 station MAE. Adding the support condition reduces acceptance to 1.40% and yields 0.856. A selector trained on relative loss can itself extrapolate into a poorly represented score configuration. A separate support condition changes that behavior without retraining the neural predictor.

The temperature fault supplies a negative control on generality. Support95 scores 0.904 under the 20 °C offset, still worse than OpenET at 0.853. Figure 3 and Table 9 report this failure alongside the wind result. VPD corruption changes no predictions in the no-VPD arm, as required by the implementation’s exact negative control.

6.3 Matched budgets limit the stronger interpretation

The fixed-threshold results motivate a post hoc ranking diagnostic with equal station-weighted acceptance budgets. Spread and support scores are divided by their respective inner 95th-percentile thresholds before pooling folds. Predicted gain is ranked from largest to smallest. The outer score distribution supplies the ranks without labels. A randomized boundary observation gives an exact acceptance budget.

Table 1. Natural-test selective correction. Each cell reports station MAE followed by the paired 95% improvement interval against OpenET, in mm day^{-1} . Proximity uses 7,923 rows, 85 stations, and 63 groups; joint uses 649 rows, 27 stations, and 24 groups. Every method retains fallback rows. Intervals are exploratory and unadjusted.

Method	Proximity / archived	Proximity / no VPD	Joint / archived	Joint / no VPD
OpenET	0.856 [0.000, 0.000]	0.856 [0.000, 0.000]	0.870 [0.000, 0.000]	0.870 [0.000, 0.000]
Full	1.099 [-0.906, 0.094]	0.835 [-0.028, 0.082]	0.766 [0.002, 0.238]	0.792 [-0.035, 0.220]
Spread95	0.817 [-0.013, 0.104]	0.832 [-0.024, 0.082]	0.776 [-0.004, 0.226]	0.786 [-0.029, 0.224]
Support95	0.818 [-0.015, 0.104]	0.838 [-0.029, 0.077]	0.763 [0.004, 0.242]	0.789 [-0.034, 0.222]
Gain	1.126 [-0.943, 0.039]	0.843 [-0.027, 0.064]	0.834 [-0.000, 0.079]	0.815 [-0.012, 0.165]
SupportGain	0.841 [-0.014, 0.048]	0.845 [-0.028, 0.062]	0.834 [-0.000, 0.079]	0.814 [-0.011, 0.164]
Uniform	0.989 [-0.464, 0.018]	0.816 [0.002, 0.093]	0.870 [0.000, 0.000]	0.835 [0.002, 0.086]
Clip	0.828 [-0.034, 0.098]	0.834 [-0.028, 0.083]	0.765 [0.003, 0.237]	0.790 [-0.033, 0.220]

We score its expected loss, rather than interpolate its two predictions. This is a transductive ranking diagnostic; it is not the deployment rule evaluated above.

At 50% acceptance on the no-VPD wind probe, Support ranks yield 1.000 expected station MAE, versus 1.023 for Spread. Their difference is $0.023 \text{ mm day}^{-1}$, with paired interval $[-0.022, 0.066]$. Thus, the apparent fixed-threshold support advantage does not establish a superior uncertainty ranking at equal coverage. This negative result is essential to the interpretation.

The Gain ranking fails more clearly in this specific probe. At the same 50% budget it scores 1.243, versus 1.000 for Support. The paired difference is $0.243 \text{ mm day}^{-1}$, with interval $[0.110, 0.412]$. On the identical unmodified rows, Gain scores 0.835 versus 0.867 for Support. That point advantage is 0.031, with interval $[-0.0003, 0.0690]$; it is not decisive evidence of natural-data superiority. The reversal identifies a failure of the fitted benefit selector under a specified covariate fault, after removing the known VPD anomaly.

The joint wind comparison does not confirm the same advantage: Gain minus Support is 0.032, with interval $[-0.105, 0.194]$. Temperature-fault comparisons are also inconclusive. Figure 4 gives the complete curves, and the saved records include every input specification and fault. No curve or acceptance budget is used to refit a model.

6.4 Cropland transfer is a separate ML target

The natural cropland subset contains 2,670 observations from 32 stations in 18 proximity groups. After VPD omission, OpenET scores 0.931 station MAE, while the full residual ensemble scores $0.997 \text{ mm day}^{-1}$. All selective alternatives have higher cropland point error than OpenET; uniform shrinkage is closest at $0.961 \text{ mm day}^{-1}$. Its paired improvement interval is $[-0.074, 0.017]$, so this comparison does not establish a cropland benefit. The later-year cropland subset contains 403 observations from nine stations in seven groups. There, no-VPD SupportGain scores 0.790 against 0.785 for OpenET, with improvement interval $[-0.035, 0.017]$. Appendix F reports every method and both input specifications. These results limit the agricultural interpretation: identifying a correction failure is useful evidence, but it does not validate a better cropland estimator. We withdraw the earlier H2-strat claim: the Phase 2 result is pooled and does not establish an OpenET benefit for croplands or irrigated croplands.

7 Discussion and limits

7.1 What the narrow result establishes

The fitted benefit selector can transfer differently from the neural predictor it controls. In the no-VPD wind probe, a correct relative-loss training objective does not prevent the selector from accepting damaging corrections. Support screening suppresses that specific failure, but much of its fixed-threshold advantage comes from greater fallback. Matching acceptance separates two questions: whether the selector rejects unfamiliar inputs and whether it ranks useful corrections better. The support-versus-disagreement ranking difference is inconclusive at the specified 50% budget. The larger Gain-versus-Support difference applies to one spatial probe and does not confirm in the smaller later-year cohort.

For agricultural deployment, selection must improve the error of the relevant crop population. The cropland results show why an aggregate mixed-land-cover score cannot supply that evidence. The present result supports diagnostics for correction behavior; it does not establish an agricultural deployment gain.

7.2 Scope of inference

All new analyses extend an inspected benchmark. The selective protocol is fixed before its run, but follows knowledge of the covariate anomaly and prior outcomes. The matched-budget analysis is specified after the fixed-threshold results and is explicitly post hoc. The bootstrap conditions on fitted models and selector ranks, omits retraining uncertainty, and has no multiplicity correction. The 62 probe groups, rather than 7,873 days, supply its main independent resampling units.

The three-seed ensemble measures initialization variation, not calibrated posterior uncertainty. The neural architecture and selectors have fixed budgets rather than exhaustive tuning. No matched implementation of a leading deferral surrogate, neural process, or GeoQ model is evaluated. The result cannot establish superiority over those methods. An independent archive and additional natural shifts are necessary to test whether the identified failure generalizes.

The controlled faults preserve labels and the satellite fallback by construction. They isolate corrupted predictor inputs; they do not simulate the physical consequences of changing wind or temperature. They do not estimate real fault frequencies. These probes do not demonstrate pure covariate shift with an unchanged conditional target law. Corrupting a mea-

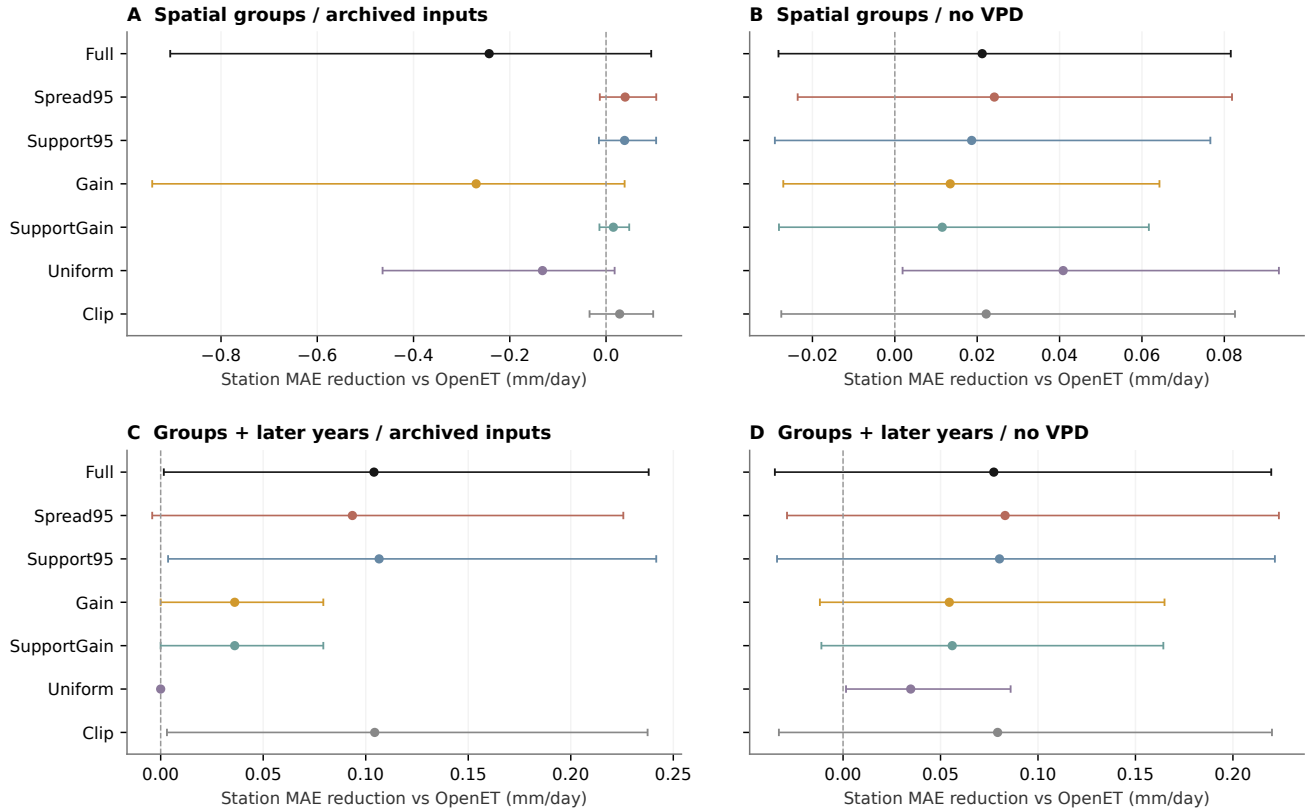


Figure 2. Paired improvement intervals for the selective methods. Positive values favor correction over unchanged OpenET. Horizontal scales differ across panels. The broad archived-input spatial interval reflects the concentrated neural failure.

sured input can change that law. Omitting VPD also does not certify the remaining sensors, energy-balance corrections, or target measurements. The instrument-level origin of the negative vapor-pressure exports remains unresolved.

The cohort selects 85 of 152 joined stations with complete covariates. This selection can favor particular sensors, climates, and land covers. The archive does not establish irrigated status for every cropland station. Available measurements at flux towers do not establish covariate availability at uninstrumented fields. The upstream satellite products may incorporate development knowledge from this archive; withholding applies to the additional MLET models.

8 Conclusion

Selective neural correction requires a comparison with the fallback’s error. In this ET archive, a cross-fitted benefit selector fails under a controlled wind-input shift even after the anomalous VPD feature is removed. Support screening reduces that failure mainly by rejecting more corrections. At equal acceptance, support does not decisively outrank ensemble disagreement, while its advantage over the fitted benefit ranking is specific to the spatial wind probe. These results identify a testable weakness in selective residual modeling and preserve the cases that do not confirm it.

Acknowledgements

I thank Meetpal S. Kukal for research mentorship and critical feedback on an earlier manuscript. I thank the OpenET and

flux-data contributors for making the source datasets available. Figure styling adapts Chen Liu’s `figures4papers` repository, with attribution and license information in the accompanying materials. The analysis and conclusions are the responsibility of the author.

Data, code, and AI assistance

The source datasets are available through the cited Zenodo records. The accompanying reproducibility package contains checksums, cohort rows, split assignments, model predictions, protocols, tests, and figure-generation scripts. The arXiv source package includes the files required to rebuild this manuscript. The repository contains the model runs, selector protocols, and saved inner and outer predictions. The numerical environment uses Python 3.13.5, NumPy 2.4.3, pandas 2.3.3, and scikit-learn 1.8.0. Codex assists with code development, data auditing, analysis, figures, and manuscript drafting. All reported experimental values come from executed scripts and saved prediction records.

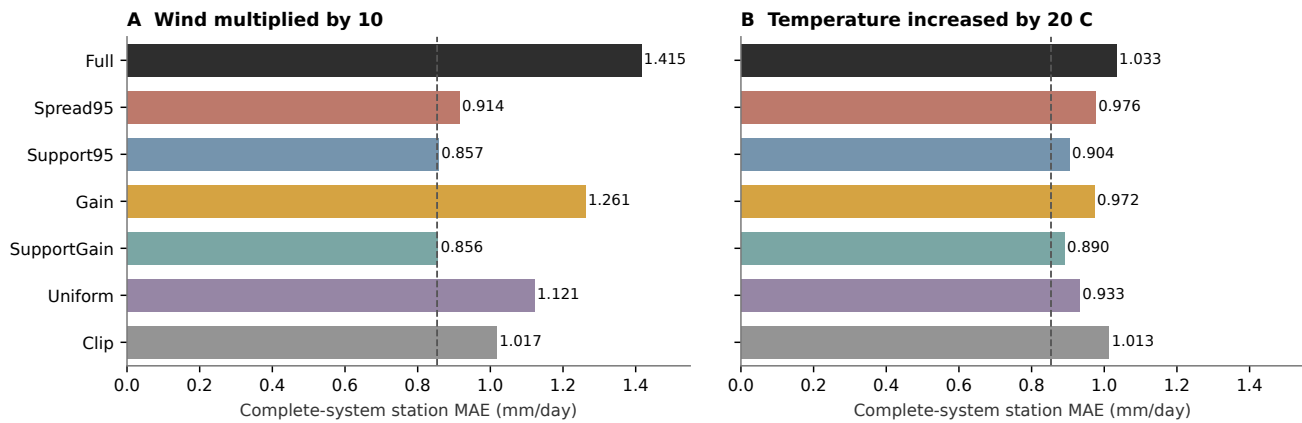


Figure 3. Paired input faults on 7,873 eligible proximity-test observations after VPD omission. The dashed line is unchanged OpenET. Each complete system includes its fallback predictions. Different acceptance rates contribute to these differences. Table 9 reports the other evaluation and the exact negative control.

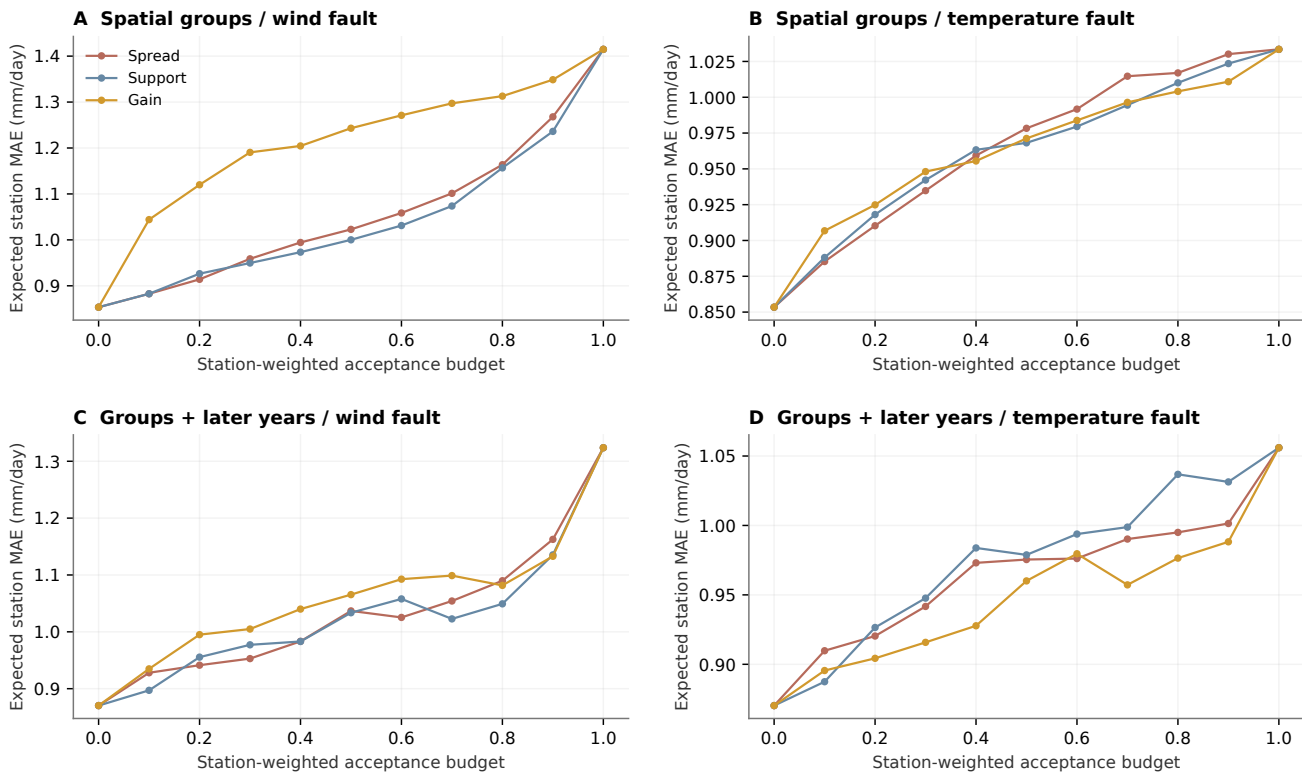


Figure 4. Expected station MAE at matched station-weighted acceptance budgets, with VPD omitted. The boundary decision is randomized to match each budget exactly. Every other rejected correction returns OpenET. These post hoc curves use label-free outer ranks; they do not establish deployable thresholds. The paired 50% comparisons appear in the text.

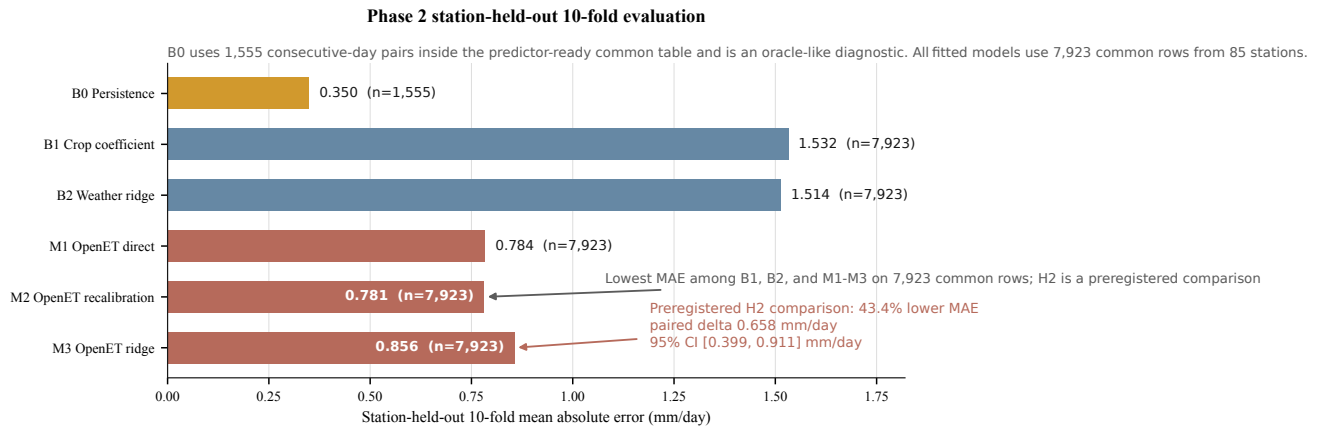


Figure 5. The original station-held-out model visual, retained unchanged. H2 (the preregistered OpenET comparison) concerns B2 weather ridge and M3 combined ridge. Its matched sample has 7,923 rows. The previous-label persistence diagnostic uses 1,555 pairs and is not a matched correction baseline.

A Original benchmark and covariate mechanisms

A.1 The earlier headline reproduces, but it compares information sets

The station split reproduces all five earlier fitted-model MAEs to numerical precision. Weather ridge scores $1.514 \text{ mm day}^{-1}$ and combined ridge scores $0.856 \text{ mm day}^{-1}$. Their pooled difference remains $0.658 \text{ mm day}^{-1}$. The original station-bootstrap interval is $[0.399, 0.911] \text{ mm day}^{-1}$. Unchanged OpenET scores $0.784 \text{ mm day}^{-1}$, while affine calibration scores $0.781 \text{ mm day}^{-1}$. Thus, the 43.4% reduction demonstrates the value of the satellite input relative to that weather baseline. It does not demonstrate a useful learned correction of OpenET.

The nonlinear weather baseline reduces station-split pooled MAE to $1.332 \text{ mm day}^{-1}$. It still trails unchanged OpenET on the same 7,923 observations. The proximity results in Table 2 likewise show the importance of the satellite baseline. No original-input learned correction has a strictly positive pri-

mary 95% interval against unchanged OpenET.

A.2 One physically inconsistent station dominates several errors

The source workbook specifies kPa for both actual vapor pressure and VPD. Among selected observations, 50 rows at six stations have negative actual vapor pressure. All 32 selected records at *mani1acotton* are in this set. Its actual vapor pressure ranges from -170.744 to -20.847 kPa . Its VPD ranges from 23.811 to 174.231 kPa . The largest VPD at the other 84 stations is 4.720 kPa . Negative actual vapor pressure is physically invalid. We cannot identify the upstream processing cause from these exported files. We therefore do not assert a unit-conversion error or invent a correction factor.

These 32 records constitute 0.404% of the selected observations. In the primary split, they contribute 79.1% of weather-ridge squared error and 26.9% of combined-ridge squared error. For the direct and residual neural ensembles, the corresponding shares are 36.9% and 74.4%. These are exact descriptive shares in the evaluated cohort, not estimates of a population prevalence. Figure 6 relates the source inconsistency to the error concentration.

At this station, the residual neural ensemble has MAE $25.321 \text{ mm day}^{-1}$. Its MAE on the other 7,891 records is $0.741 \text{ mm day}^{-1}$. That contrast concerns the already fitted predictions. Excluding the station from scoring does not establish how a model trained on a corrected archive would behave. We therefore perform explicit refits that omit VPD everywhere while retaining every target observation.

A.3 Covariate ablation removes the large neural failure

Omitting VPD reduces primary residual-network station MAE from 1.099 to $0.835 \text{ mm day}^{-1}$. Pooled MAE falls from 0.841 to $0.769 \text{ mm day}^{-1}$. At the suspect station, MAE falls from 25.321 to $1.420 \text{ mm day}^{-1}$. The direct neural model shows the same broad pattern. These refits use six inputs and 1,313 parameters, with all other neural settings unchanged.

The ablation removes valid VPD information as well as invalid values. It also changes the input dimension and corresponding neural initialization. Consequently, it is a controlled

Table 2. Primary proximity-group evaluation on 7,923 observations, 85 stations, and 63 groups. All errors and MAE differences use mm day^{-1} . Positive Δ favors the candidate over unchanged OpenET. Intervals use paired component resampling. Neural rows average three seeds.

Model	Pooled MAE	RMSE	Station MAE	Δ	95% interval
OpenET	0.784	1.066	0.856	0.000	[0.000, 0.000]
Affine calibration	0.788	1.067	0.867	-0.011	[-0.050, 0.028]
Weather ridge	1.673	4.221	2.019	-1.162	[-2.657, -0.399]
Combined ridge	0.829	1.248	0.961	-0.104	[-0.334, 0.025]
Tuned ridge	0.846	1.283	0.968	-0.111	[-0.364, 0.034]
Weather boosting	1.358	1.944	1.201	-0.345	[-0.485, -0.208]
Direct boosting	0.774	1.037	0.841	0.015	[-0.036, 0.076]
Residual boosting	0.764	1.032	0.832	0.025	[-0.018, 0.082]
Gated boosting	0.763	1.032	0.836	0.020	[-0.001, 0.047]
Direct neural	0.784	1.267	0.933	-0.076	[-0.354, 0.093]
Residual neural	0.841	1.985	1.099	-0.243	[-0.906, 0.094]

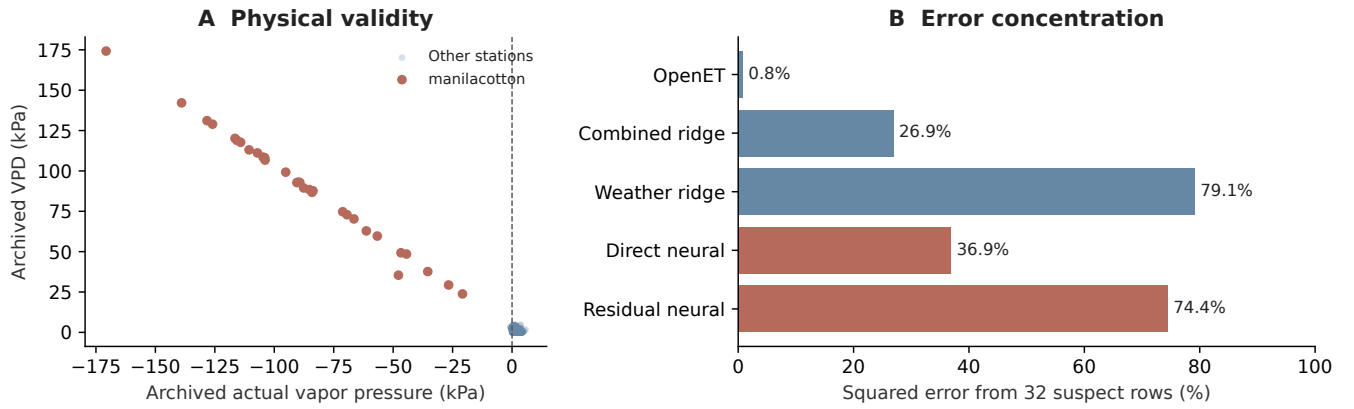


Figure 6. Input validity and error concentration in the primary split. Left: archived actual vapor pressure and VPD on selected dates. Right: squared-error shares from 32 observations at the suspect station. The source values support a physical-inconsistency diagnosis, not an identified conversion factor.

feature ablation, not a pointwise reconstruction of the true meteorology. Its value is to test whether the original conclusion survives without reliance on that input.

Combined ridge also improves when VPD is omitted, from 0.961 to 0.864 station MAE. Its remaining difference from OpenET is inconclusive. A separate input-support control clips weather features at their training 1st and 99th percentiles. Combined ridge then scores 0.866 station MAE, similar to the no-VPD result. This control retains VPD but does not reconstruct physical measurements.

Figure 7 shows the paired intervals after the failure is removed. Both no-VPD neural models remain statistically inconclusive against unchanged OpenET under proximity grouping. For the residual network, the difference is 0.021 mm day^{-1} , with interval $[-0.028, 0.082]$. Using station identifiers alone gives a larger difference of 0.052 mm day^{-1} , with interval $[0.008, 0.109]$. Thus, even after covariate ablation, the evidence for improvement depends on the treatment of nearby stations. We cannot isolate proximity dependence from the accompanying change in fold composition using one fold allocation.

A.4 Correction alignment and agricultural transfer differ

Original-input gated boosting scores 0.836 station MAE, compared with 0.856 for OpenET. Its improvement interval is $[-0.001, 0.047]$ mm day^{-1} . The no-VPD gate scores 0.831, with an exploratory improvement interval of $[0.004, 0.054]$. This small all-land-cover result should not be described as a general agricultural gain.

Across 32 cropland stations, unchanged OpenET scores 0.931 station MAE. The no-VPD gate scores 0.954, giving a difference of -0.023 with interval $[-0.044, 0.003]$. The no-VPD residual network scores 0.997 on this stratum. The all-land-cover improvement therefore does not transfer to the agricultural point estimate. Table 5 reports both populations explicitly.

An additional post hoc control fits a separate OpenET intercept and slope within each training land-cover class. It uses the global affine fit when fewer than three training observations represent a class. Test land-cover metadata selects the applicable equation; test target values never fit it. This simple control scores 0.820 station MAE in the primary split, compared with 0.856 for unchanged OpenET. Its paired improvement interval is $[-0.018, 0.087]$ mm day^{-1} . The joint result is 0.830, with improvement interval $[-0.079, 0.169]$ on 649 observa-

Table 3. Post hoc covariate sensitivity on the same primary test rows. Intervals concern station-MAE improvement over unchanged OpenET, not the difference between each pair of refits. All values use mm day^{-1} .

Model	Inputs	Pooled MAE	Station MAE	95% improvement interval
Weather ridge	Archived	1.673	2.019	[-2.657, -0.399]
Weather ridge	Omit VPD	1.439	1.366	[-0.669, -0.345]
Combined ridge	Archived	0.829	0.961	[-0.334, 0.025]
Combined ridge	Omit VPD	0.797	0.864	[-0.040, 0.031]
Direct boosting	Archived	0.774	0.841	[-0.036, 0.076]
Direct boosting	Omit VPD	0.777	0.831	[-0.018, 0.083]
Residual boosting	Archived	0.764	0.832	[-0.018, 0.082]
Residual boosting	Omit VPD	0.765	0.818	[-0.000, 0.096]
Gated boosting	Archived	0.763	0.836	[-0.001, 0.047]
Gated boosting	Omit VPD	0.762	0.831	[0.004, 0.054]
Direct neural	Archived	0.784	0.933	[-0.354, 0.093]
Direct neural	Omit VPD	0.771	0.836	[-0.029, 0.079]
Residual neural	Archived	0.841	1.099	[-0.906, 0.094]
Residual neural	Omit VPD	0.769	0.835	[-0.028, 0.082]

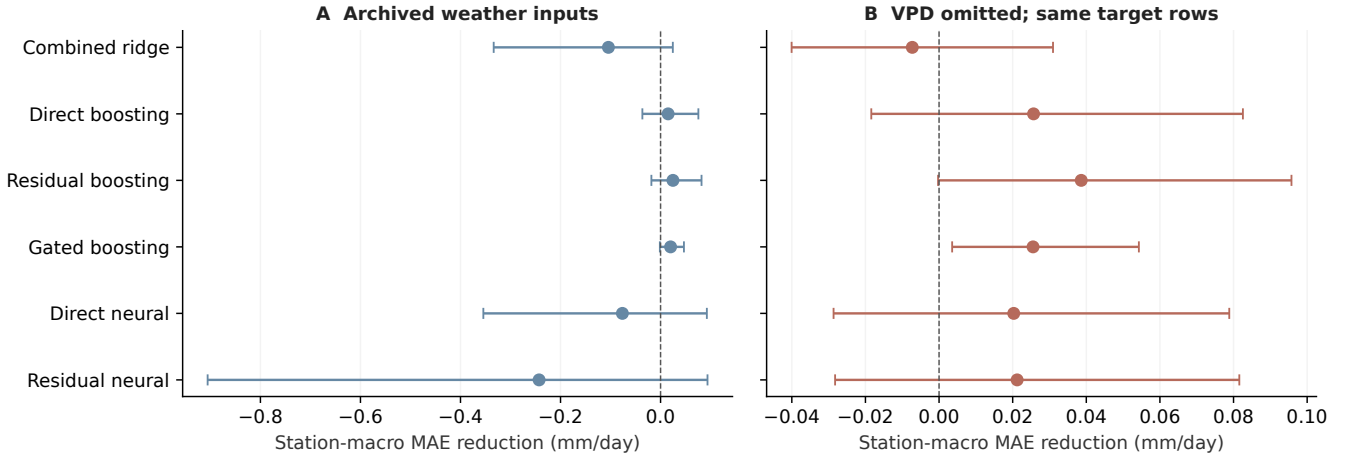


Figure 7. Paired 95% intervals for station-MAE reductions against unchanged OpenET. The primary proximity split evaluates identical target rows in both panels. Left: archived weather inputs. Right: refits without VPD. The horizontal scales differ to display the large original uncertainty and the smaller remaining effects.

tions. These intervals remain inconclusive, but the point estimates reinforce the need for simple calibration controls. This control has extra static information, so it does not isolate an architecture effect against the seven-input models.

Figure 8 diagnoses full residual boosting through Equation 4. Each point uses one station’s empirical residual alignment and correction energy. Points above the diagonal gain squared-error accuracy from a full update. Points below it lose accuracy. These test moments explain realized performance; they cannot select a deployable weight without test-label leakage. Land-cover differences in these moments describe statistical error structure, not irrigation mechanisms.

A.5 Later years expose both weighting and selection limits

Table 4 reports the joint station-and-time evaluation. Residual boosting improves station MAE from 0.870 to 0.780 mm day^{-1} . Its paired interval, $[-0.009, 0.223]$, includes zero. The same model worsens pooled MAE from 0.795 to 0.823 mm day^{-1} . These observations are compatible: stations with more records receive more influence in the pooled score.

Figure 9 shows the resulting difference in model rankings.

For the original-input residual network, the unadjusted improvement interval is $[0.002, 0.238]$ mm day^{-1} . The no-VPD version gives interval $[-0.035, 0.220]$. We retain both outcomes and do not select the favorable specification after evaluation. The small test population and multiple exploratory comparisons limit either inference.

The original-input gate selects zero correction in eight of ten outer folds. Its other weights are approximately 0.0995 and 0.0001. The inner earlier-to-later validation discourages correction, even though some outer test stations benefit from full updates. This is a negative result for the gate as a general transfer selector. An earlier validation distribution does not guarantee the correct decision under a subsequent shift.

B The algebra does not create a new representation method

Suppose a linear model uses standardized input $x \in \mathbb{R}^p$ and isotropic ridge penalty. For an orthogonal matrix Q , set $z =$

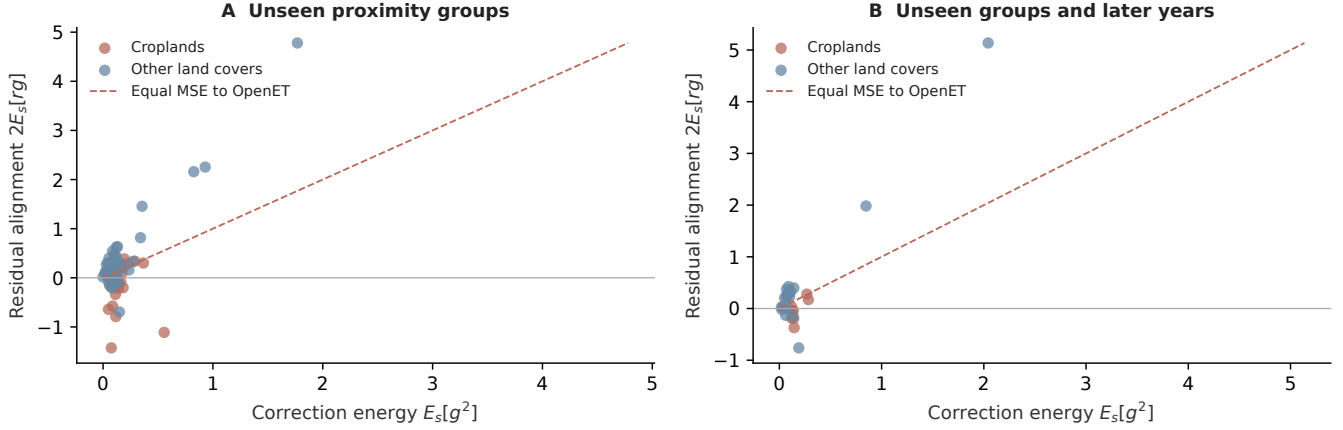


Figure 8. Exact squared-error diagnostics for original-input residual boosting. Both axes have units $(\text{mm day}^{-1})^2$. Each point is one test station; color identifies croplands versus other land covers. A full correction improves station MSE only above the diagonal. Test moments are explanatory and never select the reported gate.

Table 4. Joint transfer to withheld proximity groups in 2019–2020. All models use 649 matched observations from 27 stations in 24 groups. Training ends before 2019. Errors and differences use mm day^{-1} .

Model	Pooled MAE	Station MAE	95% station-improvement interval
OpenET	0.795	0.870	[0.000, 0.000]
Affine calibration	0.873	0.874	[-0.077, 0.068]
Weather boosting	1.799	1.417	[-0.944, -0.187]
Direct boosting	0.836	0.797	[-0.028, 0.193]
Residual boosting	0.823	0.780	[-0.009, 0.223]
Gated boosting	0.796	0.870	[-0.002, 0.000]
Direct neural	0.799	0.785	[-0.013, 0.209]
Residual neural	0.794	0.766	[0.002, 0.238]
Gated boosting (no VPD)	0.795	0.866	[-0.000, 0.009]
Direct neural (no VPD)	0.851	0.796	[-0.027, 0.203]
Residual neural (no VPD)	0.849	0.792	[-0.035, 0.220]

$Q^\top x$ and $\theta = Q^\top \beta$. Then $\theta^\top z = \beta^\top x$ and $\|\theta\|_2^2 = \|\beta\|_2^2$. The prediction and ridge objective are unchanged under this rotation. A numerical test verifies held-out prediction equivalence within 10^{-11} relative and 10^{-12} absolute tolerance. Thus, simply rotating linear features into orthogonal components cannot supply an improvement here. Additional constraints, nonlinear predictors, or a changed training objective would require separate controlled experiments. This observation prevents an unsupported attribution of novelty to factorization terminology.

For a mixture of station populations, the moments in Equation 4 also mix:

$$A = \sum_s w_s A_s, \quad B = \sum_s w_s B_s. \quad (12)$$

Changing w_s can change both the optimal correction weight and the sign of a full correction’s benefit. Pooled scoring uses $w_s = n_s/N$; station-macro scoring uses $w_s = 1/S$. This provides a direct explanation for the weighting sensitivity without requiring a change in predictions. It is an algebraic identity, not a new theorem about distribution shift.

C Cropland and other land covers

The proximity cropland subset contains 18 groups; its other-land-cover subset contains 45 groups. The joint cropland subset contains seven groups. Its small number of independent resampling units limits generalization. Models are trained across land covers; these are evaluation strata, not separate fitted crop models.

D Optimization seeds

All networks complete the specified 120 epochs. Iteration-limit warnings are retained in the run records. They identify termination at the declared budget, not a demonstrated converged optimum. The results characterize these trained models under that budget.

E Compute and reproducibility

The original-input experiment takes 93.45 seconds end to end in the recorded local run. The tree/ridge sensitivity takes 5.46 seconds; the neural sensitivity takes 82.10 seconds. These are single-run wall times, not latency benchmarks with independent repetitions. The recorded platform is macOS 26.6.2 on ARM, with the numerical thread limit set to one. Paid training-compute expenditure is USD 0; electricity cost is not

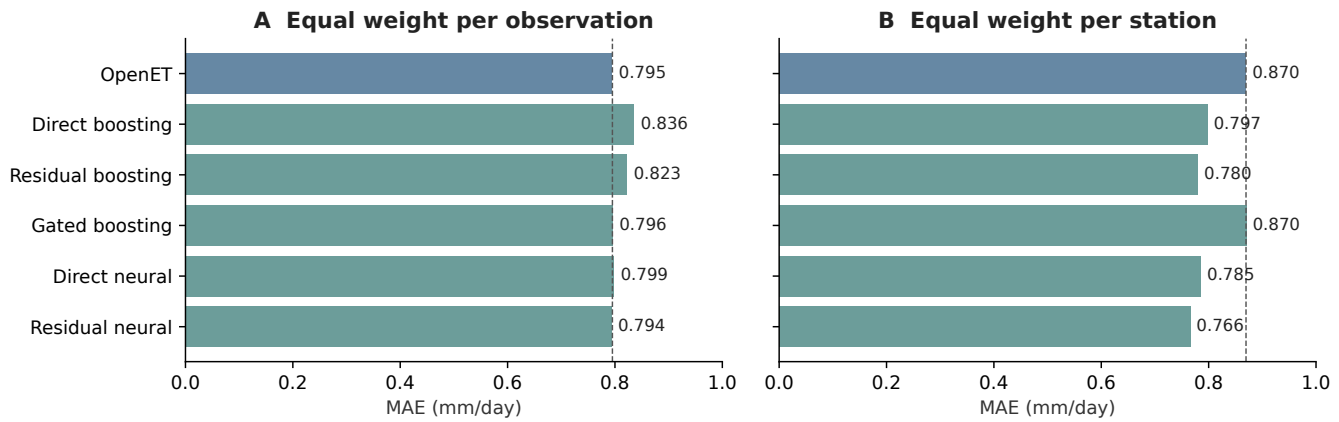


Figure 9. The weighting rule changes the joint-transfer conclusion on identical observations. The dashed line marks unchanged OpenET. Residual boosting worsens pooled MAE but improves the station-weighted point estimate. Table 4 supplies paired uncertainty intervals.

Table 5. Explicit target populations after VPD omission. OpenET is unchanged; the other models omit VPD. The interval measures station-MAE improvement against OpenET, in mm day^{-1} . Rounded endpoints near zero must not be interpreted as definitive significance.

Split	Stratum	Sites	Rows	Model	Station MAE	95% interval
Proximity	Crop	32	2670	OpenET	0.931	[0.000, 0.000]
Proximity	Crop	32	2670	Gated boosting	0.954	[-0.044, 0.003]
Proximity	Crop	32	2670	Residual neural	0.997	[-0.121, -0.000]
Proximity	Other	53	5253	OpenET	0.811	[0.000, 0.000]
Proximity	Other	53	5253	Gated boosting	0.756	[0.026, 0.086]
Proximity	Other	53	5253	Residual neural	0.738	[0.012, 0.139]
Joint	Crop	9	403	OpenET	0.785	[0.000, 0.000]
Joint	Crop	9	403	Gated boosting	0.785	[-0.004, 0.004]
Joint	Crop	9	403	Residual neural	0.916	[-0.200, -0.044]
Joint	Other	18	246	OpenET	0.912	[0.000, 0.000]
Joint	Other	18	246	Gated boosting	0.907	[0.000, 0.013]
Joint	Other	18	246	Residual neural	0.730	[0.031, 0.356]

measured. Downloads and manuscript production are outside these timings.

The original source snapshot is MLET revision 9b922ad. New scripts and protocols have separate SHA-256 receipts. The receipts retain the full revision and both source-archive MD5 values. The reproducibility package also records SHA-256 values for both archives and the rebuilt cohort.

To reproduce the experiments, acquire the two archives and verify their recorded hashes. Run the main audit, the covariate sensitivity, and the neural sensitivity in that order. Run the land-cover control with the saved outer partitions. Completed result files are protected against silent overwrite. The figure builder reads saved predictions and does not refit the models. All source humidity flags remain in the exported audit table. No repaired humidity values are substituted into the archive.

The selective experiment adds 480 neural fits: three seeds in three inner folds and one outer fit, across 40 outer configurations. Its recorded single-run wall time is 161.98 seconds on the same platform, with one numerical thread. This includes selector fitting, prediction, output writing, and reporting; it excludes literature work and manuscript production. All 40 outer neural ensembles reproduce the earlier saved predictions within relative and absolute tolerance 10^{-10} . No ensemble is

rerun to select a favorable seed.

Table 6. Station-macro MAE for every neural seed, in mm day^{-1} . The main tables use predictions averaged across seeds, not the average of these errors. No seed is selected for its test performance.

Split	Model	20260713	20260714	20260715
Station	Direct neural	1.373	0.830	1.154
Station	Residual neural	1.668	1.017	1.887
Station	Direct neural (no VPD)	0.812	0.815	0.828
Station	Residual neural (no VPD)	0.810	0.812	0.813
Proximity	Direct neural	1.337	1.183	1.373
Proximity	Residual neural	1.463	0.843	1.022
Proximity	Direct neural (no VPD)	0.850	0.842	0.845
Proximity	Residual neural (no VPD)	0.853	0.839	0.839
Joint	Direct neural	0.764	0.796	0.823
Joint	Residual neural	0.767	0.796	0.759
Joint	Direct neural (no VPD)	0.782	0.823	0.804
Joint	Residual neural (no VPD)	0.795	0.804	0.801

Table 7. Measured computation for the ten primary outer fits per model. Entries are median [25th, 75th percentile], in milliseconds. Neural rows sum all three seed costs within each fold. Prediction times cover each entire fold batch; batch sizes vary. Times exclude downloads, inner selection, reporting, and figure rendering.

Model	Fit time (ms)	Batch prediction (ms)
Affine calibration	0.33 [0.33, 0.39]	0.041 [0.040, 0.042]
Combined ridge	0.43 [0.43, 0.46]	0.043 [0.039, 0.049]
Direct boosting	33.18 [32.55, 34.10]	1.388 [1.147, 1.693]
Direct neural	1458.24 [1406.12, 1477.05]	0.415 [0.393, 0.458]
Residual boosting	32.19 [31.48, 32.51]	1.292 [1.149, 1.610]
Residual neural	1473.38 [1415.05, 1535.03]	0.444 [0.395, 0.504]
Tuned ridge	0.43 [0.42, 0.45]	0.043 [0.041, 0.047]
Weather boosting	30.66 [29.91, 31.11]	1.417 [1.124, 1.577]
Weather ridge	0.42 [0.42, 0.46]	0.046 [0.043, 0.047]

F Selective results by population and fault

Table 8. Cropland station MAE and paired 95% improvement intervals against OpenET, in mm day^{-1} . Proximity includes 2,670 rows, 32 stations, and 18 groups. Joint includes 403 rows, nine stations, and seven groups. Models train across land covers; this table reports evaluation strata.

Method	Proximity / archived	Proximity / no VPD	Joint / archived	Joint / no VPD
OpenET	0.931 [0.000, 0.000]	0.931 [0.000, 0.000]	0.785 [0.000, 0.000]	0.785 [0.000, 0.000]
Full	1.739 [-2.952, -0.001]	0.997 [-0.121, -0.000]	0.876 [-0.151, -0.036]	0.916 [-0.200, -0.044]
Spread95	0.980 [-0.112, 0.016]	0.991 [-0.116, 0.004]	0.878 [-0.153, -0.036]	0.900 [-0.182, -0.023]
Support95	0.984 [-0.116, 0.015]	0.996 [-0.120, -0.001]	0.873 [-0.149, -0.031]	0.909 [-0.200, -0.023]
Gain	1.717 [-2.929, -0.009]	0.989 [-0.105, -0.001]	0.779 [-0.013, 0.027]	0.796 [-0.038, 0.008]
SupportGain	0.958 [-0.049, 0.005]	0.989 [-0.105, -0.000]	0.779 [-0.013, 0.027]	0.790 [-0.035, 0.017]
Uniform	1.304 [-1.430, 0.000]	0.961 [-0.074, 0.017]	0.785 [0.000, 0.000]	0.796 [-0.028, 0.008]
Clip	1.017 [-0.185, 0.002]	0.995 [-0.121, 0.003]	0.875 [-0.150, -0.035]	0.914 [-0.201, -0.037]

Table 9. Complete-system station MAE after VPD omission, in mm day^{-1} . All fault columns use the same eligible rows as their clean column: 7,873 rows at 84 stations for proximity, and 646 rows at 27 stations for joint. The VPD fault is an exact negative control. These paired probes do not estimate fault prevalence.

Method	Proximity groups				Groups and later years			
	Clean	VPD $\times 10$	Wind $\times 10$	$T + 20$	Clean	VPD $\times 10$	Wind $\times 10$	$T + 20$
OpenET	0.853	0.853	0.853	0.853	0.870	0.870	0.870	0.870
Full	0.828	0.828	1.415	1.033	0.793	0.793	1.324	1.056
Spread95	0.826	0.826	0.914	0.976	0.787	0.787	0.954	0.941
Support95	0.831	0.831	0.857	0.904	0.791	0.791	0.874	0.928
Gain	0.838	0.838	1.261	0.972	0.816	0.816	1.041	0.928
SupportGain	0.840	0.840	0.856	0.890	0.814	0.814	0.871	0.882
Uniform	0.810	0.810	1.121	0.933	0.836	0.836	0.882	0.869
Clip	0.827	0.827	1.017	1.013	0.791	0.791	0.970	0.973

All archived-input fault results, per-seed predictions, selector acceptance rates, and ranking diagnostics appear in the saved numerical package. The paired 95% intervals use proximity components and condition on the fitted predictions. The matched-budget curves keep their learned scores fixed during resampling.

G Original evidence architecture

Actual evapotranspiration is the measured crop-and-soil flux used for this paper’s ML target. The reference evapotranspiration (ET_o) calculation describes a standardized surface and remains a separate physical calculation ([ASCE Task Committee on Standardization of Reference Evapotranspiration, 2005](#)). The selective models operate on the historical actual-ET branch below. The earlier reference-ET_o branch retains its source and validation constraints. The earlier OpenET information-value result is reproduced in Appendix A. Adding OpenET to weather ridge reduces pooled MAE from 1.514 to 0.856 mm day⁻¹ on the original station split. Unchanged OpenET already scores 0.784 on those rows. This demands a stronger objective for the present study: determine whether, and when, a correction improves the existing satellite estimate.

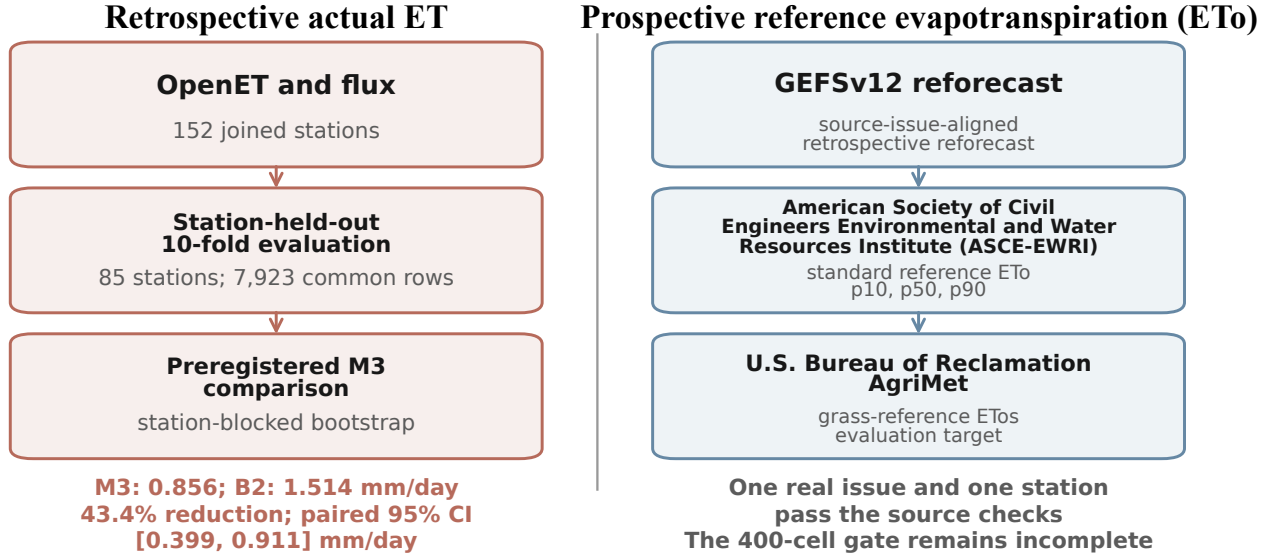


Figure 10. The original MLET evidence architecture, retained unchanged. The left branch supplies 7,923 station-days from 85 stations for retrospective actual-ET evaluation. The right branch contains a separate reference-ET_o protocol and one diagnostic with 20 targets. The new selective neural experiments concern the left branch.

G.1 Physical target definitions

The original MLET quantities retain the following definitions and evidence requirements.

Table 10. Target ontology and claim boundary.

Layer or label	Quantity	Time and source	Allowed claim
ET_corr	Energy-balance-corrected actual ET	Historical flux label	Retrospective daily actual-ET prediction
eto_mm	Standardized short-reference ET _o	Future local date from issue-time GEFS	Regional reference-ET _o outlook after the frozen hindcast gate
AgriMet ETos	Published grass-reference ET	Later station observation; converted from inches per day	Independent reference-ET _o evaluation target
potential_et_c_mm	K_c ET _o under ample-water assumptions	Conditional future projection	Conditional crop ET only
eta_analysis_mm	Satellite actual-ET analysis	Delayed historical observation	Analysis comparison only
Conditional ET _a scenarios	Named water and soil assumptions	Future scenario	Scenario evidence only

H The retained reference-ETo outlook

This separate branch predicts standardized reference ET, not the actual-ET target studied above. The American Society of Civil Engineers Environmental and Water Resources Institute (ASCE-EWRI) specifies the reference calculation ([ASCE Task Committee on Standardization of Reference Evapotranspiration, 2005](#)). The National Oceanic and Atmospheric Administration (NOAA) supplies weather forecasts from its Global Ensemble Forecast System version 12 (GEFSv12). The U.S. Bureau of Reclamation (USBR) AgriMet archive supplies published grass-reference evapotranspiration (ETos) observations. The BOII (the Boise, Idaho AgriMet weather-station identifier) case is a retrospective reforecast diagnostic for July 3, 2019 at 00Z (00:00 UTC). Its recorded archive retrieval at 2026-08-04T18:08:54.243122Z is a later retrieval timestamp; it does not prove operational availability at issue time or original publication time. The case has one issue, one station, and $n = 20$ daily targets. Its source receipts use SHA-256 (a 256-bit secure hash). The frozen gate requires support across all lead-season-fold cells; this case does not meet that requirement or establish general forecast skill. The selective regression results do not change the outlook release gate. The displayed p10-to-p90 interval is an uncalibrated ensemble quantile band.

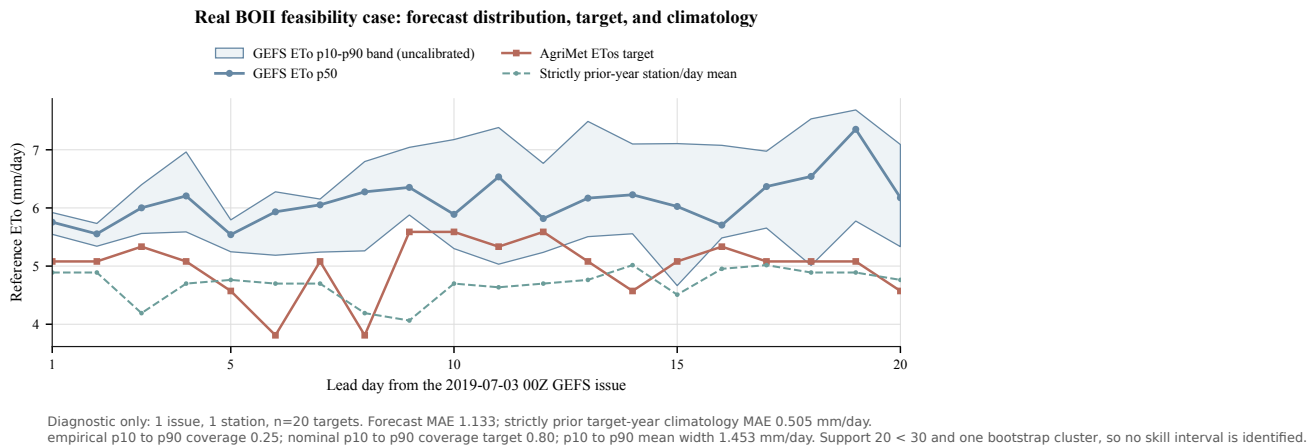


Figure 11. One real BOII feasibility case. The GEFS member distribution gives p10, p50, and p90 reference ET_0 for leads 1 through 20. The independent target is AgriMet ETos. The baseline uses the same station and target day of year with observations from strictly prior calendar years. The scope is one issue, one station, and $n = 20$ targets. It is diagnostic only.

The prior-station, target-day climatology has MAE $0.505 \text{ mm day}^{-1}$. The baseline-minus-forecast MAE difference is $-0.628 \text{ mm day}^{-1}$, so forecast MAE is higher. Forecast p50 RMSE is $1.295 \text{ mm day}^{-1}$, with bias $1.133 \text{ mm day}^{-1}$. The p10-to-p90 band has empirical coverage 0.25 against nominal coverage 0.80. These values have no sampling interval and only 1 bootstrap cluster. Each occupied lead-season-fold cell has one target, below the required minimum of 30. The result is a negative feasibility diagnostic, not general forecast skill.

References

- ASCE Task Committee on Standardization of Reference Evapotranspiration. *The ASCE Standardized Reference Evapotranspiration Equation*. American Society of Civil Engineers, 2005. doi: 10.1061/9780784408056.
- Taoyong Cui, Zhongyao Wang, Xinyue Xu, et al. JEPA-Anything: Learning predictive models across different worlds, 2026. URL <https://arxiv.org/abs/2609.20800>.
- Antoine de Mathelin, Francois Deheeger, Mathilde Mougeot, and Nicolas Vayatis. Deep anti-regularized ensembles provide reliable out-of-distribution uncertainty quantification. *arXiv:2304.04042*, 2023. URL <https://arxiv.org/abs/2304.04042>.
- Aleksei Dobrokhoto, Oliver Lopez, Ludmila Kozyreva, and Daria Mukhina. A machine learning residual correction framework for enhancing physical-based evapotranspiration estimates over croplands. *European Journal of Remote Sensing*, 59:2695909, 2026. doi: 10.1080/22797254.2026.2695909.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillcrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. doi: 10.1038/s41586-025-08744-2. URL <https://arxiv.org/abs/2301.04104>. Associated preprint: arXiv:2301.04104.
- Tristan E. M. Hascoet, Victor Pellet, and Filipe Aires. Learning evapotranspiration dataset corrections from water cycle closure supervision. In *NeurIPS Workshop on Tackling Climate Change with Machine Learning*, 2022. URL <https://www.climatechange.ai/papers/neurips2022/23>.
- Junfeng Hu, Yuxuan Liang, Zhencheng Fan, Hongyang Chen, Yu Zheng, and Roger Zimmermann. Graph neural processes for spatio-temporal extrapolation. *arXiv:2305.18719*, 2023. URL <https://arxiv.org/abs/2305.18719>.
- Meetpal S. Kukal, Richard G. Allen, Ayse Kilic, et al. Artificial intelligence/machine learning estimation of FAO and ASCE standardized reference ET erodes the physical basis and intended purpose of ET standardization. *Agricultural Water Management*, 333: 110657, 2026. doi: 10.1016/j.agwat.2026.110657.
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *arXiv:1612.01474*, 2016. URL <https://arxiv.org/abs/1612.01474>.
- Anqi Mao, Mehryar Mohri, and Yutao Zhong. Regression with multi-expert deferral. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 34738–34759, 2024. URL <https://proceedings.mlr.press/v235/mao24d.html>.
- Hanna Meyer and Edzer Pebesma. Predicting into unknown space? estimating the area of applicability of spatial prediction models. *arXiv:2005.07939*, 2020. URL <https://arxiv.org/abs/2005.07939>.
- Khoa Nguyen, Daniel Serino, Aviral Prakash, and Marc Klasky. GeoQ: Geometry-aware conditional quantile error estimation for scientific surrogate models. *arXiv:2608.21652*, 2026. URL <https://arxiv.org/abs/2608.21652>.
- OpenET. Accuracy and known issues, 2026. URL <https://etdata.org/accuracy-known-issues/>. Accessed September 22, 2026.
- David R. Roberts, Volker Bahn, Simone Ciuti, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40:913–929, 2017. doi: 10.1111/ecog.02881.
- Aleksei Rozanov, Samikshya Subedi, Vasudha Sharma, and Bryan C. Runck. Knowledge-guided machine learning models to upscale evapotranspiration in the U.S. Midwest, 2025. URL <https://arxiv.org/abs/2510.11505>.
- Haiyang Shi. Extrapolability improvement of machine learning-based evapotranspiration models via domain-adversarial neural networks, 2024. URL <https://arxiv.org/abs/2406.00805>.
- Anna Sokol, Nuno Moniz, and Nitesh Chawla. Conformalized selective regression. *arXiv:2402.16300*, 2024. URL <https://arxiv.org/abs/2402.16300>.
- John M. Volk, Justin L. Huntington, Forrest S. Melton, et al. Assessing the accuracy of OpenET satellite-based evapotranspiration data to support water resource and land management applications. *Nature Water*, 2:193–205, 2024. doi: 10.1038/s44221-023-00181-7.
- John M. Volk et al. Post-processed data and graphical tools for a CONUS-wide eddy flux evapotranspiration dataset, 2023a. URL <https://zenodo.org/records/7636781>.
- John M. Volk et al. OpenET model data for assessing the accuracy of OpenET satellite-based evapotranspiration data, 2023b. URL <https://zenodo.org/records/10119477>.
- Jiaxing Wei, Shaomin Liu, Lisheng Song, et al. Development and validation of physically constrained machine learning for improving remote sensing-based evapotranspiration estimation. *Remote Sensing of Environment*, 2026. doi: 10.1016/j.rse.2026.115460.
- Keyulu Xu, Jingling Li, Mozhi Zhang, Simon S. Du, Ken-ichi Kawarabayashi, and Stefanie Jegelka. How neural networks extrapolate: From feedforward to graph neural networks. *arXiv:2009.11848*, 2020. URL <https://arxiv.org/abs/2009.11848>.